

EPO ACADEMY TRAINING MATERIAL

PATSTAT Explorer

User Guide

Patent analysis on the EPO Technology Intelligence Platform.
Tutorial walkthrough using a Siemens applicant example.

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mtc.berlin · Built for the EPO Academy and the PATLIB TIP Working Group, 2025–2026.

What PATSTAT Explorer does

Inputs, outputs, scope.

INPUT

Applicant name, CPC code, or plain text

Find a company by any name variant (Siemens AG, Siemens Ltd, Siemens Corp).

Search a technology by CPC code (H01M, G06N) or natural language (battery, neural network).

Compare countries by ISO codes (DE, US, CN, JP, KR).

Search patent titles and abstracts by keyword.

OUTPUT

Charts, tables, networks, and CSV

Filing trend, top jurisdictions, top CPC fields per applicant.

Co-applicant network graph and citation Sankey diagram.

CPC section co-occurrence heatmap.

CSV export on every result table.

Bookmarkable URLs that share the full analysis state.

Launching from TIP

Three steps. First run takes 1–2 minutes.

1

Download the notebook

Open the [GitHub repo](#) and download PATSTAT_Explorer.ipynb (or use File then Save Link As on the raw link).

2

Upload to TIP JupyterHub

Open tip.epo.org, sign in, and upload the notebook to your home directory.

3

Run the cell

Open the notebook, click into the code cell, and run it. A clickable link appears once the app is ready.

Note: the notebook starts three local services on your TIP environment (schema helper, Python API, web app). They run inside JupyterHub and stop when the server is restarted.

Four entry points

Pick the one that matches your question.

✓ PATSTAT Explorer is running

PATSTAT Explorer open →

Node v24.4.1 · Port 52080

■ Stop PATSTAT Explorer

Applicant Search

Start with a company name. Group name variants and analyse the consolidated portfolio.

Technology Search

Start with a CPC code or a description. See the top filers in that area.

Country Comparison

Compare filing trends across a set of countries, optionally filtered to a CPC code.

AI Query

Ask a question in plain English. Claude writes and runs the SQL. Bring your own API key.

Applicant search

Find every PATSTAT name variant of one organisation.

Applicant Search

Search by name, then click to analyse a single applicant -or select multiple checkboxes to group name variants (e.g. "Siemens AG" + "Siemens Ltd") into a single analysis.

Search

Filing years: –

200 results

3082ms · Click a name to analyse · Use checkboxes to group name variants

Auto-suggest

All

None

CSV

☐

Siemens Aktiengesellschaft

DE · 73.666 families

suggest

☐

Siemens AG

DE · 46.100 families

suggest

☐

SIEMENS AKTIENGESELLSCHAFT

DE · 38.175 families

suggest

☐

Siemens AG

· 34.495 families

suggest

☐

SIEMENS & HALSKE AKTIENGESELLSCHAFT

· 19.898 families

suggest

☐

SIEMENS-SCHUCKERTWERKE AKTIENGESELLSCHAFT

· 19.083 families

suggest

☐

SIEMENS AKTIENGESELLSCHAFT

· 16.419 families

suggest

☐

SIEMENS AG, 80333 MUENCHEN, DE

DE · 12.864 families

suggest

☐

SIEMENS AG, 1000 BERLIN UND 8000 MUENCHEN, DE

· 10.772 families

suggest

☐

SIEMENS-SCHUCKERTWERKE AKT.-GES.

· 9.826 families

suggest

Use checkboxes to select multiple names for consolidation, or click a name to analyse directly.

What you do

- Type a name fragment, for example SIEMENS, in the search box.
- Adjust the filing year range if you only need a recent window.
- Sort the results by family count to see the largest variants first.

What you see

A table of distinct applicant names PATSTAT holds, with country code and family count for each. Variants like SIEMENS AG, SIEMENS LTD, SIEMENS CORP appear as separate rows.

Why it matters

PATSTAT does not deduplicate organisation names. Without consolidation, a portfolio split across name variants looks smaller than it is.

PATSTAT Explorer

5 · Tutorial

Name consolidation

Group variants into one analysis entity.

200 results146 selected3082ms · Click a name to analyse · Use checkboxes to group name variants

Auto-suggestAllNoneCSV

☒ Siemens AktiengesellschaftPARENTDE · 73.666 families

[suggest](#)[use as name](#)

☒ Siemens AGDE · 46.100 families

[suggest](#)[use as name](#)

☒ SIEMENS AKTIENGESELLSCHAFTDE · 38.175 families

[suggest](#)[use as name](#)

☒ Siemens AG· 34.495 families

[suggest](#)[use as name](#)

☒ SIEMENS & HALSKE AKTIENGESELLSCHAFT· 19.898 families

[suggest](#)[use as name](#)

☐ SIEMENS-SCHUCKERTWERKE AKTIENGESELLSCHAFT· 19.083 families

[suggest](#)

☒ SIEMENS AKTIENGESELLSCHAFT· 16.419 families

[suggest](#)[use as name](#)

☒ SIEMENS AG, 80333 MUENCHEN, DEDE · 12.864 families

[suggest](#)[use as name](#)

☒ SIEMENS AG, 1000 BERLIN UND 8000 MUENCHEN, DE· 10.772 families

[suggest](#)[use as name](#)

☐ SIEMENS-SCHUCKERTWERKE AKT.-GES.· 9.826 families

[suggest](#)

☒ SIEMENS AG, 1000 BERLIN UND 8000 MUENCHEN· 9.268 families

[suggest](#)[use as name](#)

☐ SIEMENS-SCHUCKERTWERKE AKTIENGESELLSCHAFTDE · 7.417 families

[suggest](#)

☒ SIEMENS & HALSKE AKT.-GES.· 5.989 families

[suggest](#)[use as name](#)

☒ SIEMENS & HALSKE AKTIENGESELLSCHAFTDE · 5.808 families

[suggest](#)[use as name](#)

☐ SIEMENS AG

[suggest](#)[use as name](#)

Consolidate & Analyse

Names grouped146

Total families361.456

Display name for the grouped analysis:

☒ Siemens Aktiengesellschaft (most filings)

☐ Custom name

Preview Trend

Analyse Consolidated

Selected Names

Siemens Aktiengesellschaft73.666

Siemens AG46.100

SIEMENS AKTIENGESELLSCHAFT38.175

Siemens AG34.495

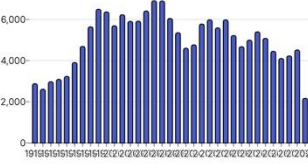
SIEMENS & HALSKE AKTIENGESELLSCHAFT19.898

SIEMENS AKTIENGESELLSCHAFT16.419

SIEMENS AG, 80333 MUENCHEN, DE12.864

Consolidated Trend

CSV



What you do

- Tick the rows that belong to the same organisation.
- Pick a parent name from the suggestions or type a custom one.
- Confirm the selection. The preview chart shows the merged filing trend.

What you see

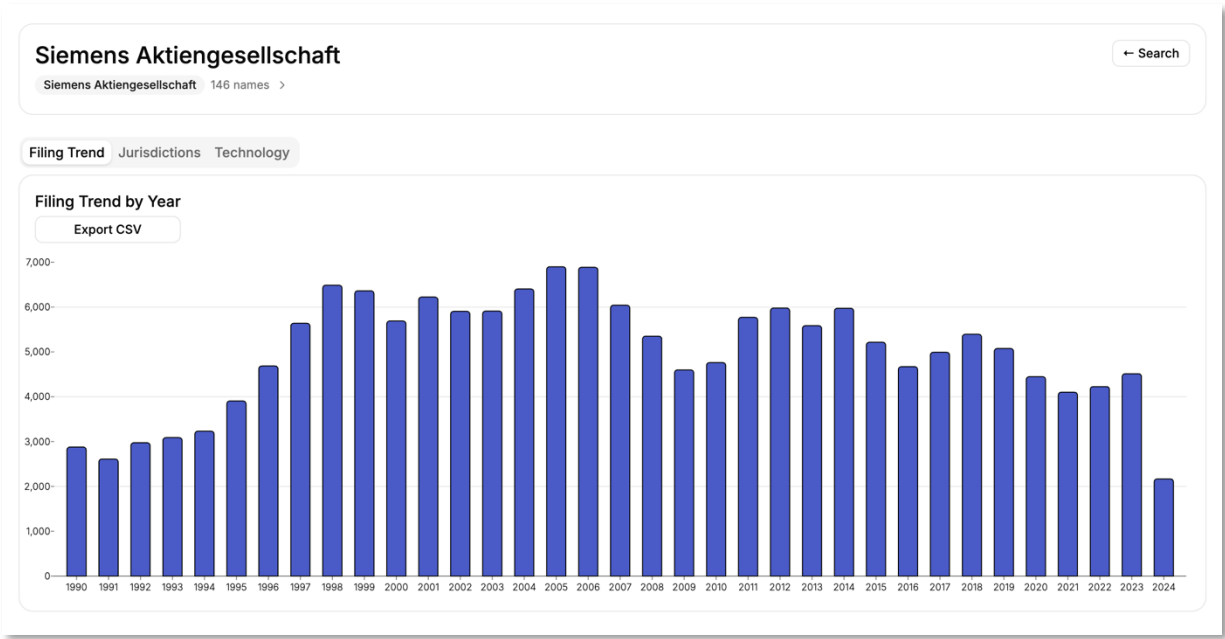
A combined family count, a preview filing trend across the selected variants, and a context bar at the top of the page that follows you into every other view.

Result

All downstream views (Overview, Network, Citations, CPC Map) operate on the consolidated set and show the parent label as the entity name.

Deep-dive: Overview

Filing trend, jurisdictions, and top technology fields.



Three panels

- Filing trend by year. Identifies growth periods, plateaus, and recent shifts.
- Top jurisdictions. Where the organisation files, ranked by family count.
- Top CPC technology fields. Which sub-classes dominate the portfolio.

Tabs at the top

Overview, Network, Citations, CPC Map. Each tab keeps the same applicant context. The URL updates as you switch tabs and is safe to share.

Export

Each panel has a CSV download for the underlying data.

Network: who files together

[illegible]

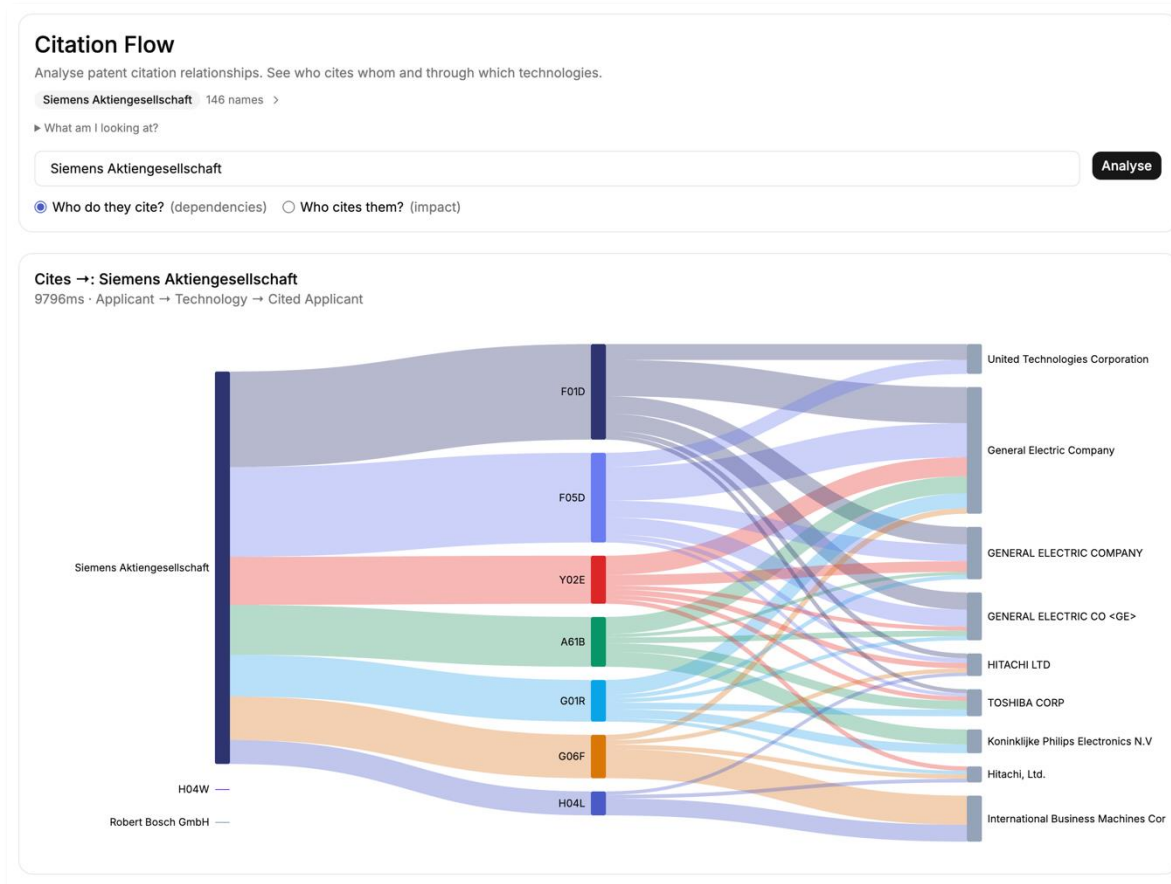
- The centre node is the consolidated applicant.
- Linked nodes are co-applicants on shared patent families.
- Edge weight encodes the number of shared families.

- Drag a node to reposition it.
- Scroll to zoom, drag the background to pan.
- Hover for the exact shared-family count.

Reveals research consortia, joint ventures, and inventor-employer relationships that are not obvious from filings alone.

Citations: who cites whom

Forward and backward citation flows by technology.



Two directions

- Forward: which organisations and technology fields cite the focal applicant.
- Backward: which organisations and technology fields the focal applicant cites.

How to read the Sankey

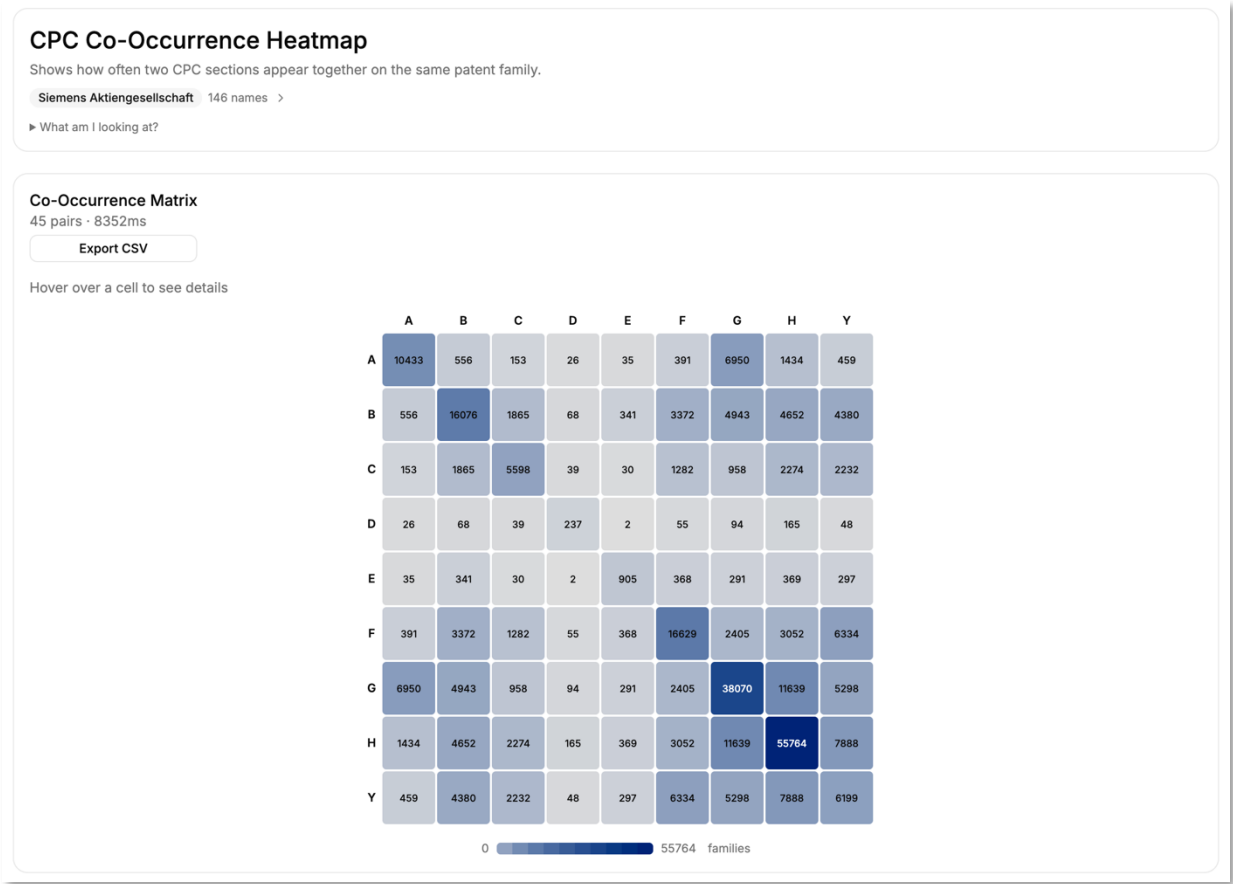
Left column is one party, right column is the other. Band width encodes the citation count between them.

Use case

Forward citations indicate downstream impact. Backward citations indicate the knowledge base the applicant builds on. Reading them together shows technology dependencies in both directions.

CPC Map: technology overlap

How CPC sections co-occur on the same patent families.



What it shows

Heatmap of pairwise co-occurrences across the 8 main CPC sections (A to H plus Y for emerging tech). A dark cell at row H column G means many of the applicant's families carry both Electricity and Physics sections.

Diagonal vs off-diagonal

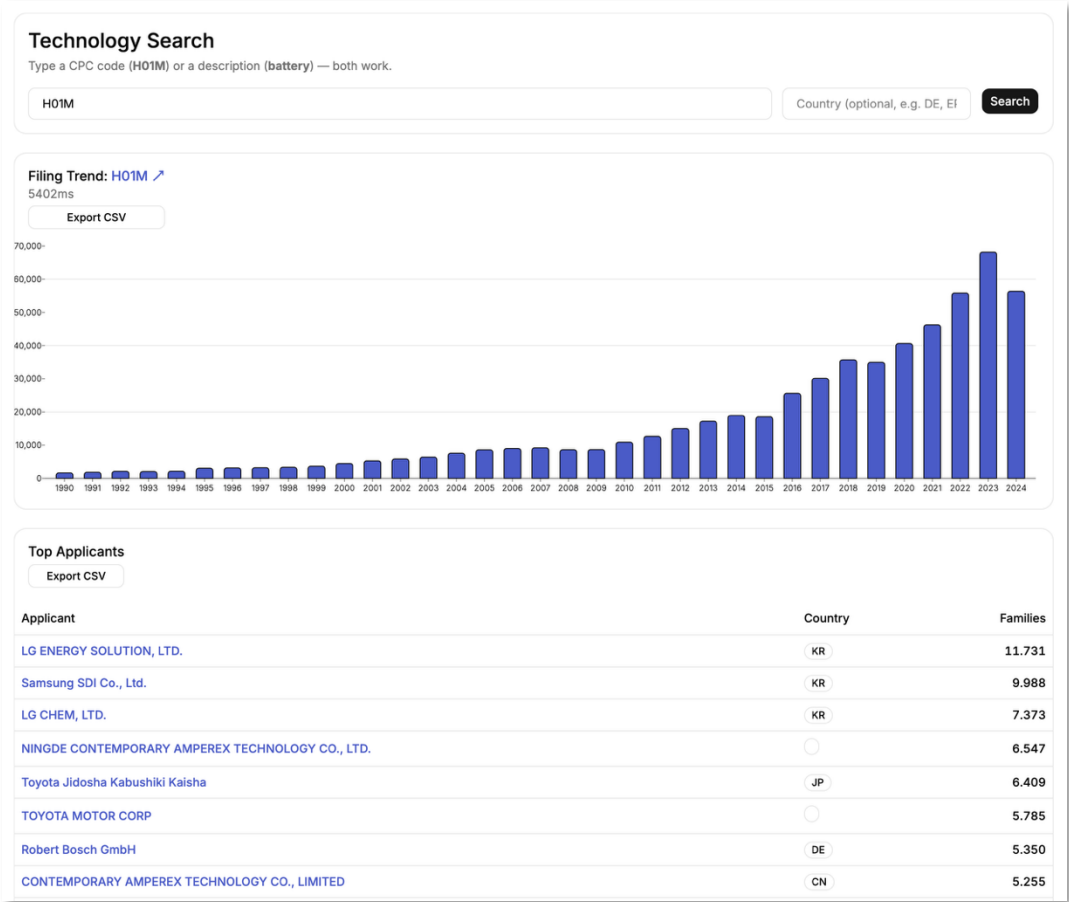
- Diagonal: families filed in a single section. High values indicate concentration.
- Off-diagonal: cross-sectional families. High values indicate interdisciplinary work.

Use case

Spot the technology profile of an organisation in one glance: specialist (one bright cell), generalist (spread), or interdisciplinary (off-diagonal).

Technology search

Start from a CPC code or a plain-language description.



Two input modes

- CPC code: H01M, G06N, A61K. Section, class, subclass, group, or subgroup.
- Plain text: battery, neural network, vaccine. The app suggests matching CPC codes.

Result

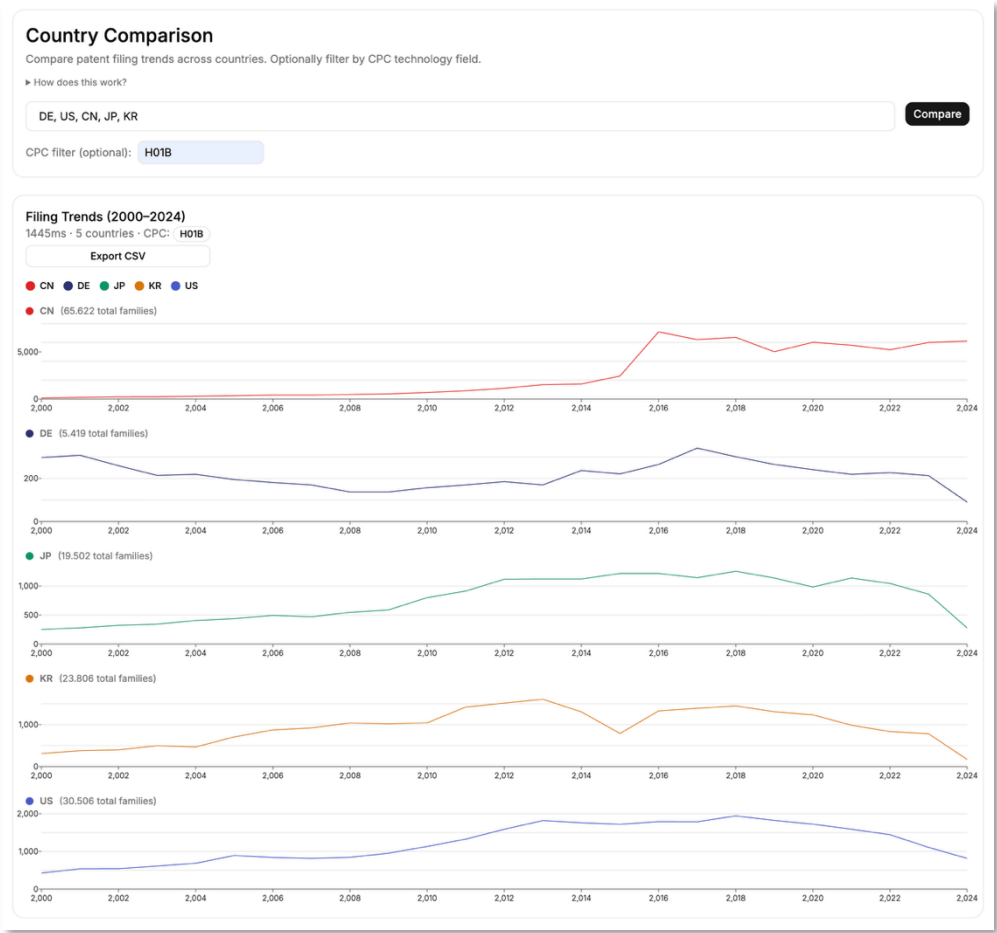
Top applicants in the selected technology, with country code and family count. Optional country filter narrows the result set.

Click through

Click any applicant to enter the same Deep-dive used in the Applicant Search workflow. The technology context follows along.

Country comparison

Filing trends across a set of jurisdictions.



Inputs

- Country list as ISO codes, comma-separated. Default is DE, US, CN, JP, KR.
- Optional CPC filter to restrict the comparison to one technology.

Output

One line per country, family count by filing year. Each country has a distinct colour. CSV export available.

Use case

Quick view of geographic shifts in patent activity, either overall or for a specific technology field.

Bookmarkable URLs

Every analysis state is encoded in the URL.

Why this matters

Selected applicants, name consolidations, technology filters, and active tab all live in the URL query string. Copy the address bar to send a colleague the exact view you are looking at, with no setup on their side.

Example

```
.../applicant?names=SIEMENS%20AG,SIEMENS%20LTD&label=Siemens&tab=network
```

Tips

- Bookmark a deep-dive view to come back to the same consolidation later.
- Paste a URL into a report or email; the receiver lands on the same charts.
- URLs survive a notebook restart. The app rebuilds the view from the parameters.

Export and limits

Where data goes, and what to keep in mind.

CSV EXPORT

Available on every result table and chart panel.

Files are UTF-8 with a header row. Numbers are unformatted, ready for pivot tables or downstream scripts.

File names include the analysis context (applicant label, year range, tab) so downloads stay self-explanatory.

THINGS TO KEEP IN MIND

PATSTAT release. The data reflects the PATSTAT version available on TIP at run time. Recent filings may not yet be present.

Query latency. Network and citation views run heavier queries; allow a few seconds.

Family counts. All counts are at DOCDB family level unless noted otherwise.

Name consolidation is manual. The app suggests groupings; you confirm. Always check before drawing conclusions.

AI Query

Ask a question. Claude writes the SQL.

Use this when the built-in views do not cover your question. The AI Query interface uses the PATSTAT MCP server to give Claude the schema, then runs SQL on your behalf and returns rows.

Natural LanguageSQL

Top 10 applicants in solid-state battery patents (CPC H01M10) since 2015

Generated SQL query.

```
SELECT
  p.person_name,
  COUNT(DISTINCT a.docdb_family_id) as patent_families
FROM t1s201_appln a
JOIN t1s224_appln_cpc cpc ON a.appln_id = cpc.appln_id
JOIN t1s207_pers_appln pa ON a.appln_id = pa.appln_id
JOIN t1s206_person p ON pa.person_id = p.person_id
WHERE a.appln_filing_year >= 2015
AND pa.appln_seq_nr > 0
AND (cpc.cpc_class_symbol LIKE 'H01M 10/0562%'
      OR cpc.cpc_class_symbol LIKE 'H01M 10/0565%')
GROUP BY p.person_name
ORDER BY patent_families DESC
LIMIT 10
```

RunEdit in SQL tab

MCPmtc.berlin PATSTAT MCP Server

```
list_tables
get_table_schema ('table_name': 't1s201_appln')
get_table_schema ('table_name': 't1s224_appln_cpc')
get_table_schema ('table_name': 't1s207_pers_appln')
get_table_schema ('table_name': 't1s206_person')
search_tables ('keyword': 'H01M10')
execute_query ('query': 'SELECT * FROM t1s_cpc_hierarchy WHERE symbol LIKE 'H01M10%' LIMIT 10')
execute_query ('query': 'SELECT * FROM t1s_cpc_hierarchy WHERE symbol LIKE 'H01M%' LIMIT 10')
execute_query ('query': 'SELECT symbol, symbol_patstat, title_en FROM t1s_cpc_hierarchy WHERE symbol LIKE 'H01M0010%' AND title_en LIKE '%solid%' LIMIT 10')
execute_query ('query': 'SELECT symbol, symbol_patstat, title_en FROM t1s_cpc_hierarchy WHERE symbol LIKE 'H01M0010%' ORDER BY symbol LIMIT 20')
execute_query ('query': 'SELECT symbol, symbol_patstat, title_en FROM t1s_cpc_hierarchy WHERE symbol LIKE 'H01M0010%' AND title_en LIKE '%electrolyte%' LIMIT 10')
execute_query ('query': 'SELECT symbol, symbol_patstat, title_en FROM t1s_cpc_hierarchy WHERE symbol LIKE 'H01M0010%' AND (title_en LIKE '%solid%' OR symbol_patstat LIKE '%0562%' OR symbol_patstat LIKE '%0565%') LIMIT 15')
execute_query ('query': 'SELECT (p.person_name, COUNT(DISTINCT a.docdb_family_id) as patent_families) FROM t1s201_appln a JOIN t1s224_appln_cpc cpc ON a.appln_id = cpc.appln_id JOIN t1s207_pers_appln pa ON a.appln_id = pa.appln_id JOIN t1s206_person p ON pa.person_id = p.person_id WHERE a.appln_filing_year >= 2015 AND pa.appln_seq_nr > 0 AND (cpc.cpc_class_symbol LIKE 'H01M 10/0562%' OR cpc.cpc_class_symbol LIKE 'H01M 10/0565%') GROUP BY p.person_name ORDER BY patent_families DESC LIMIT 10')
```

e.g. Find all PEM electrolysis patents from European applicants since 2018 with at least 5 citations

SETUP

Open Settings, paste an Anthropic API key. The key is stored in your TIP session only and never sent to the repo.

HOW IT WORKS

- You ask in English. The agent inspects the schema via MCP.
- It drafts SQL, runs it against PATSTAT on BigQuery.
- Each step is visible. Switch to SQL mode to edit the generated query directly.

Note: AI Query is best for one-off questions. For repeatable analyses, prefer the dedicated views.

Help and credits

Where to look, who to contact.

LINKS

Source repository

github.com/mtcberlin/epo-tip-svelte-patstat-explorer

PATSTAT MCP server

github.com/mtcberlin/mtc-patstat-mcp-lite

EPO Technology Intelligence Platform

tip.epo.org

EPO Academy

epo.org/en/learning

CREDITS

Concept

Arne Krueger, mtc.berlin

Architecture

Matze Schmidbauer, mtc.berlin

Inspiration

Chris Toft

Built for

EPO Academy and the PATLIB TIP Working Group, 2025–2026

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